

Aliah University

End Semester Examination (Even Semester)-2026

Subject Name: Cyber Security and Cryptography

Full Marks: 80

Subject Code: CSAUGPC31

Time: 3 hrs.

Group-A

1. Answer any ten(10) question-choose the right option(any 10) 1X10=10

- I. Which of the following is a symmetric key algorithm?
a) RSA ☒ b) DES c) ECC d) Diffie-Hellman
- II. The process of converting plaintext into ciphertext is called
a) Decryption ☒ b) Encryption c) Hashing d) Authentication
- III. Which attack aims to make a service unavailable?
a) Replay attack b) MITM attack ☒ c) DoS attack d) Passive attack
- IV. In RSA, the public key consists of:
a) (p, q) b) (d, n) ☒ c) (e, n) d) ($\phi(n)$, e)
- V. Which technique rearranges characters without changing them?
a) Substitution b) Transposition c) Hashing d) Encoding
- VI. SSL is used for:
a) File storage ☒ b) Secure communication
c) Compression d) Routing
- VII. A digital signature provides:
a) Confidentiality only b) Integrity only
☒ c) Authentication and integrity d) Compression
- VIII. Which is NOT a type of authentication?
a) Password b) Token c) Biometric ☒ d) Firewall
- IX. IDEA algorithm belongs to:
a) Asymmetric cryptography ☒ b) Symmetric cryptography
c) Hash functions d) Authentication protocols
- X. SQL Injection is an example of:
a) Physical attack b) Network attack c) Web attack d) Hardware attack
- XI. Key loggers are used to:
a) Encrypt data ☒ b) Capture keystrokes c) Block traffic d) Generate keys
- XII. Block cipher processes data in:
a) Bit by bit ☒ b) Fixed-size blocks
c) Variable length streams d) Random format

- (Answer any four)
2. Differentiate between monoalphabetic and polyalphabetic ciphers with examples.
 3. Explain the working of RSA algorithm with a suitable example.
 4. Define and differentiate between block cipher and stream cipher.
 5. Discuss active and passive attacks with examples.
 6. Explain Digital Signature and Message Digest concepts.
 7. Describe SSL protocol and its importance in network security.
 8. Explain cybercrimes and tools such as key loggers, spyware, and SQL injection.

4X10=40

Group-C

(Answer any four (4) questions)

9. Describe the DES algorithm with a neat diagram and explain its working steps.
10. Explain the IDEA algorithm in detail with structure and operations.
11. Discuss public key cryptography (RSA) with key generation, encryption, and decryption.
12. Explain Internet Security Protocols and Authentication techniques (Password, Token, and Biometric).
13. Describe Man-in-the-Middle attack and its prevention techniques.
14. Explain Cyber Security mechanisms including network, web, cloud, and wireless security.